PAPER_669: UQFF Waveform Comparison — GW150914

Author: Daniel T. Murphy **Subtitle:** UQFF-modified strain h_UQFF vs LIGO GW150914 observation. Inspiral chirp mass, frequency evolution, and phase shift derived. **Module:** UQFFComparedToGW150914
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Abstract

We compute the UQFF-modified gravitational wave strain h_UQFF for the GW150914 binary black hole merger and compare to LIGO observation. UQFF introduces a ~10% strain suppression and a small phase shift controlled by κ and f_TRZ.

1. GW150914 Parameters

- m1 = 36 M⊙, m2 = 29 M⊙
- d = 410 Mpc
- f_peak = 150 Hz, h_obs ≈ 10^{-21}

2. Chirp Mass

$$\mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \approx 28.3\, M_\odot$$

3. Standard Strain

$$h_{GR}(f) = \frac{4}{d}\left(\frac{G\mathcal{M}_c}{c^2}\right)^{5/3} (\pi f)^{2/3}/c$$

4. UQFF Modifications

$$S_{SCm}(f) = \exp(-\rho_{SCm}\,\lambda_{GW}), \quad \lambda_{GW} = c/f$$

$$h_{UQFF}(f) = h_{GR}(f) \cdot (1 - f_{TRZ}) \cdot S_{SCm}(f) \cdot \exp\left(-\frac{U_m\,\omega}{c^2}\right)$$

5. Phase Shift

$$\phi_{UQFF}(t) = 2\pi f t + \kappa\, f_{TRZ}\, t$$

Small but potentially measurable with future detectors (Einstein Telescope).

6. Frequency Evolution

$$\frac{df}{dt} = \frac{96}{5}\pi^{8/3}\frac{(G\mathcal{M}_c)^{5/3}}{c^5} f^{11/3}$$

7. C++ Module

`UQFFComparedToGW150914.h` / `.cpp` — Session 172 CP4 #253 — `UQFFComparedToGW150914Calculator`

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export) ### §A.1 Sector Classification

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This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`). ### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm): $\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{BH}})(\partial^\mu \phi_{\text{BH}}) - V(\phi_{\text{BH}}) + \mathcal{L}_{\text{cosmo}}$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and: $V(\phi_{\text{BH}}) = \frac{1}{2}m^2\phi_{\text{BH}}^2 + \frac{\lambda}{4!}\phi_{\text{BH}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{BH}}$ ### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{BH}}} = R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R + \rho_{\mathrm{vac},[\mathrm{SCm}]} g_{\mu\nu} + F_{U\_Bi\_i}/r^2 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\mathrm{BH}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of7 motion

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.170 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 29, \quad n_{\text{channel}} = 20/26$$

Since $p_{\text{DVP}} = 29$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^6 M_BH/M_⊙ yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.170	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 29$	✓ Resonant
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	✓ Full 26D projection

Framework	Canonical Value	This Paper	Status
κ decay	5.0×10^{-4} day^{-1}	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.